

Proxy-Master Pipeline — Flow Diagrams

Ring Proposer → Quick Estimator → Shortcut Solver → Ring Exclude, with wavelength postprocessing

This document shows the full workflow as flow graphs. The glossary defines the symbols used in the figures. Every decision diamond states its full test in place — there is no separate decision-node appendix. Colour encodes module ownership throughout: **blue** = Ring Proposer, **gold** = Quick Estimator, **green** = Shortcut Solver, **red** = wavelength postprocessing, grey = control logic.

1 Glossary

Quantities

Symbol	Produced by	Meaning and role
LB_k	Proxy master	Lower bound at the start of round k . Non-decreasing across rounds.
LB_{k+1}	Proxy master	Lower bound after this round's cuts and the Re-Solve.
W_{proxy}	Proxy master	Master objective; numerically the current LB .
$\widehat{W}(R)$	Quick Estimator	Fast estimate for ring R . Ranking only — neither a bound nor a proof input.
$W_{\text{true}}(R)$	Shortcut Solver	True achievable worst-case value on the fixed ring R .
W^*	Shortcut Solver (min)	Incumbent: best W_{true} found so far; an upper bound on the optimum.
gap	bookkeeping	$W^* - LB$. > 0 unproven; ≈ 0 certified optimal; < 0 inconsistent lower bound.
ObjBound	Shortcut Solver MIP	Solver lower bound on $W_{\text{true}}(R)$ during Phase 2; used for the dual-safe abort.
λ_{needed}	Wavelength post	Number of wavelengths from the conflict-graph colouring. Postprocess only.
$\delta(i, j)$	—	Manhattan distance $ x_i - x_j + y_i - y_j $.
L_c	Shortcut Solver Phase 1	True routed length of candidate c ; always $L_c \geq \delta(c)$.
ε	—	Numerical tolerance used in all bound comparisons.
$\text{Adm}(q)$	Wavelength post	Admissible routes for demand q : enumerated routes of length $\leq W^*$.

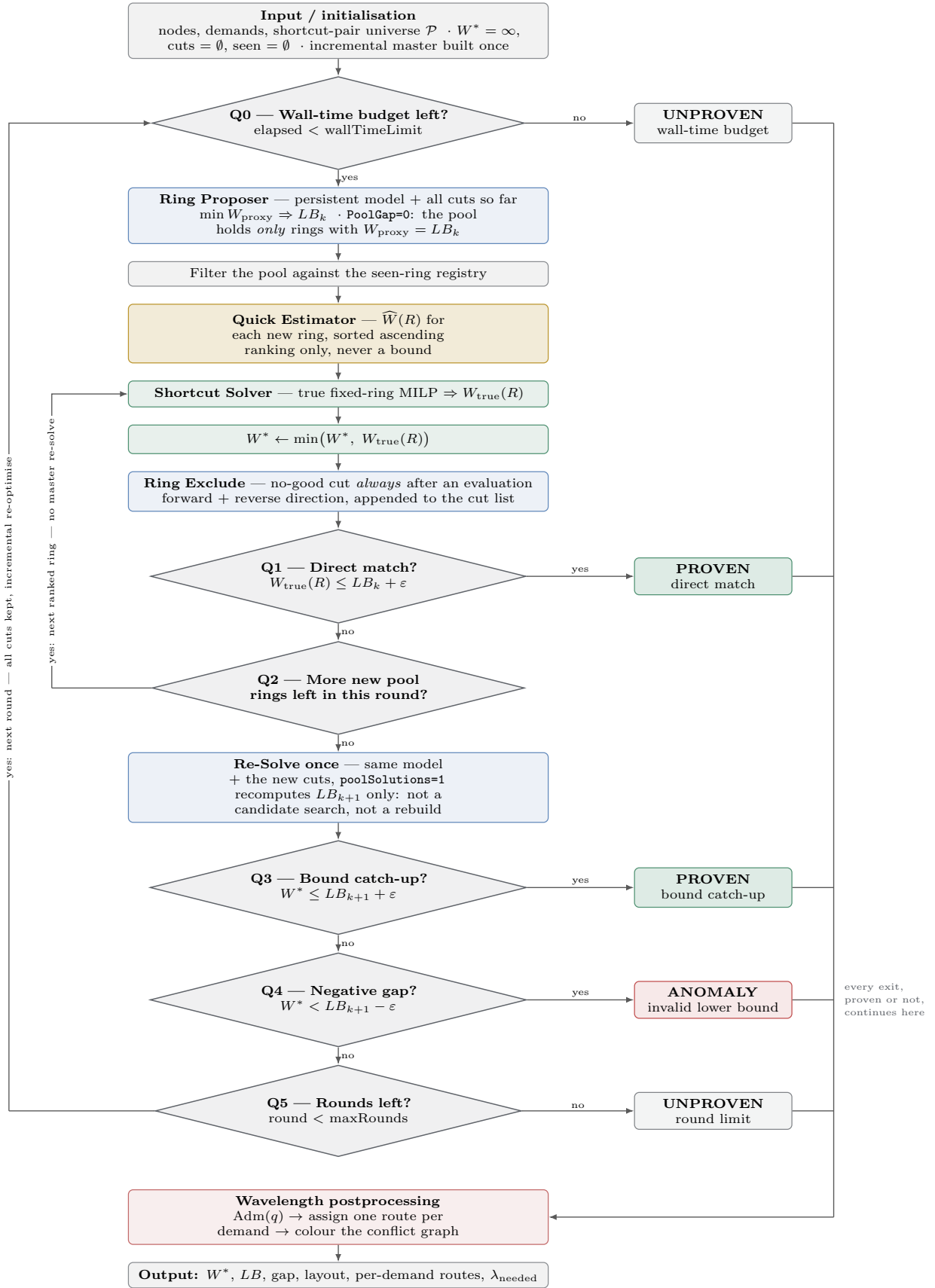
Model variables

Symbol	Meaning
b_{ij}	Binary: the ring tour goes directly $i \rightarrow j$ (directed).
r_{ij}	Binary: L-shape of an active ring edge ($0 = \text{HV}$, $1 = \text{VH}$).
z_c	Binary: shortcut c selected — an abstract endpoint pair in the master, a real routed path in the Shortcut Solver.
f	Continuous flow on ring and shortcut edges.
W	Continuous worst-case path length (the objective).

Abbreviations

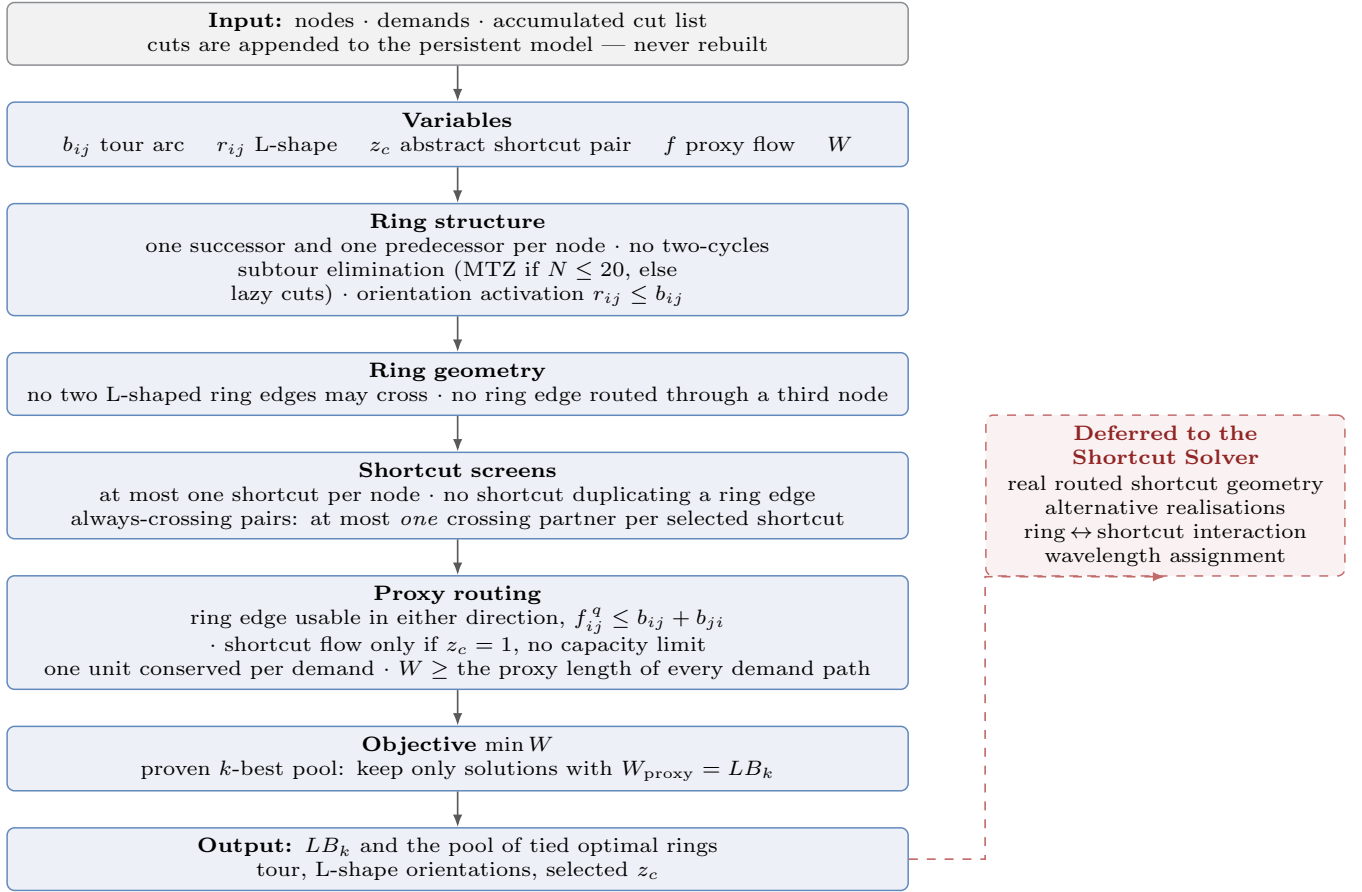
Term	Meaning
Ring Proposer / Quick Estimator / Shortcut Solver / Ring Exclude	The four steps of one round, in the order they run: $M \rightarrow E \rightarrow V \rightarrow C$.
LB / UB	Lower bound / upper bound.
MTZ	Subtour elimination via order variables ($N \leq 20$); larger N uses lazy cuts.
HV / VH	The two rectilinear L-shape orientations: horizontal-then-vertical, vertical-then-horizontal.
CW / CCW	Clockwise / counter-clockwise: two independent directional ring waveguides.
No-good cut	Constraint excluding one evaluated ring in both tour directions.
Direct match	$W_{\text{true}}(R) \leq LB_k + \varepsilon$.
Round catch-up	$W^* \leq LB_{k+1} + \varepsilon$, tested after the Re-Solve.
Seen-ring registry	Set of already-evaluated ring signatures, so a ring is never verified twice.

2 Diagram 1 — Overall pipeline



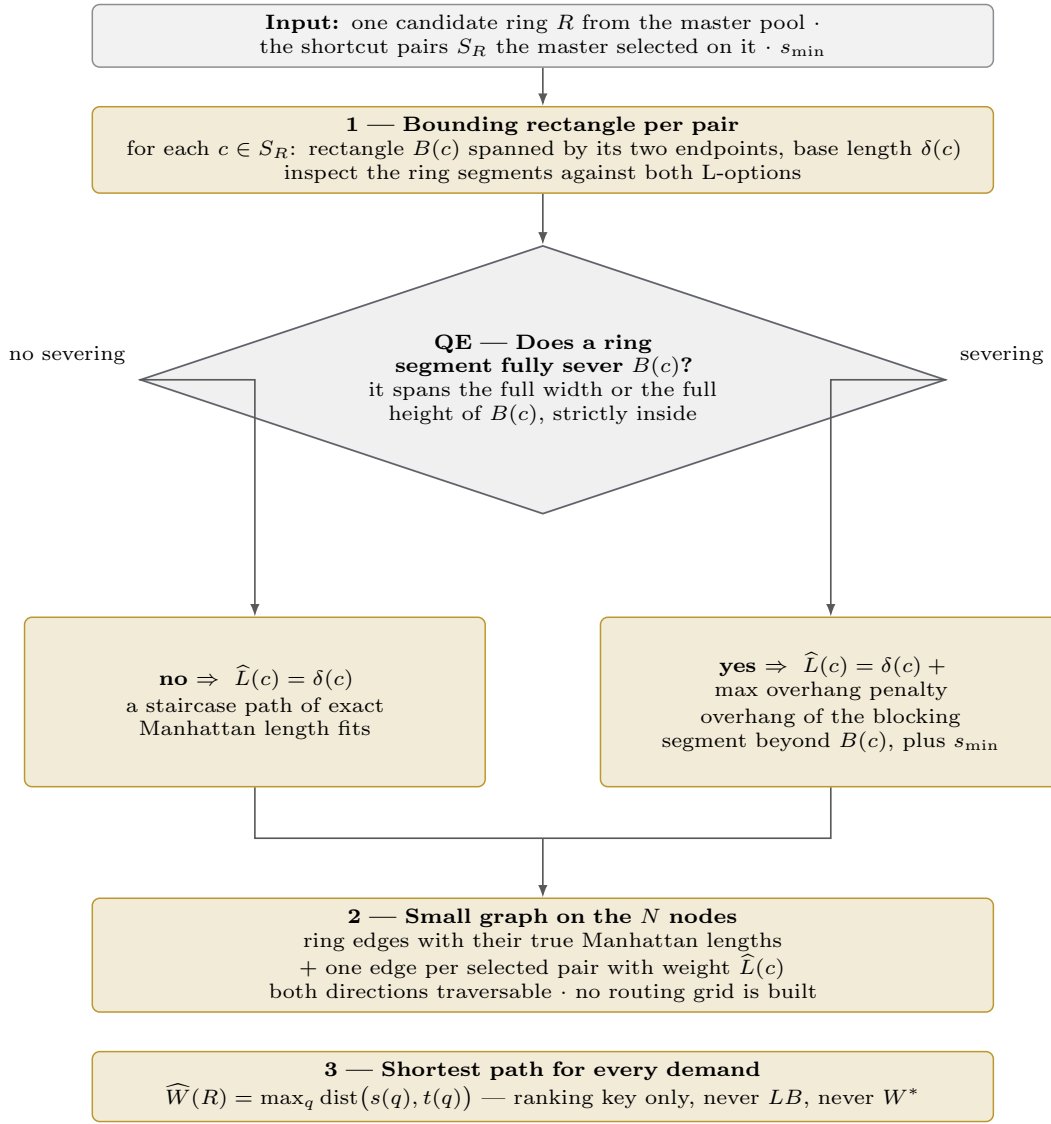
How to read this. The inner loop at **Q2** verifies several tied pool rings without re-solving the master. The **Re-Solve** that follows it is bound-only: one solve, no pool, its sole purpose is to refresh LB_{k+1} for **Q3** and **Q4**. The outer loop at **Q5** starts the next round on the same persistent model with all cuts kept — an incremental re-optimize, never a rebuild. Outcome boxes are coded by result: green proven, grey unproven, red anomaly. Wavelength postprocessing runs exactly once, after the loop ends, whichever exit was taken.

3 Diagram 2 — Inside the Ring Proposer



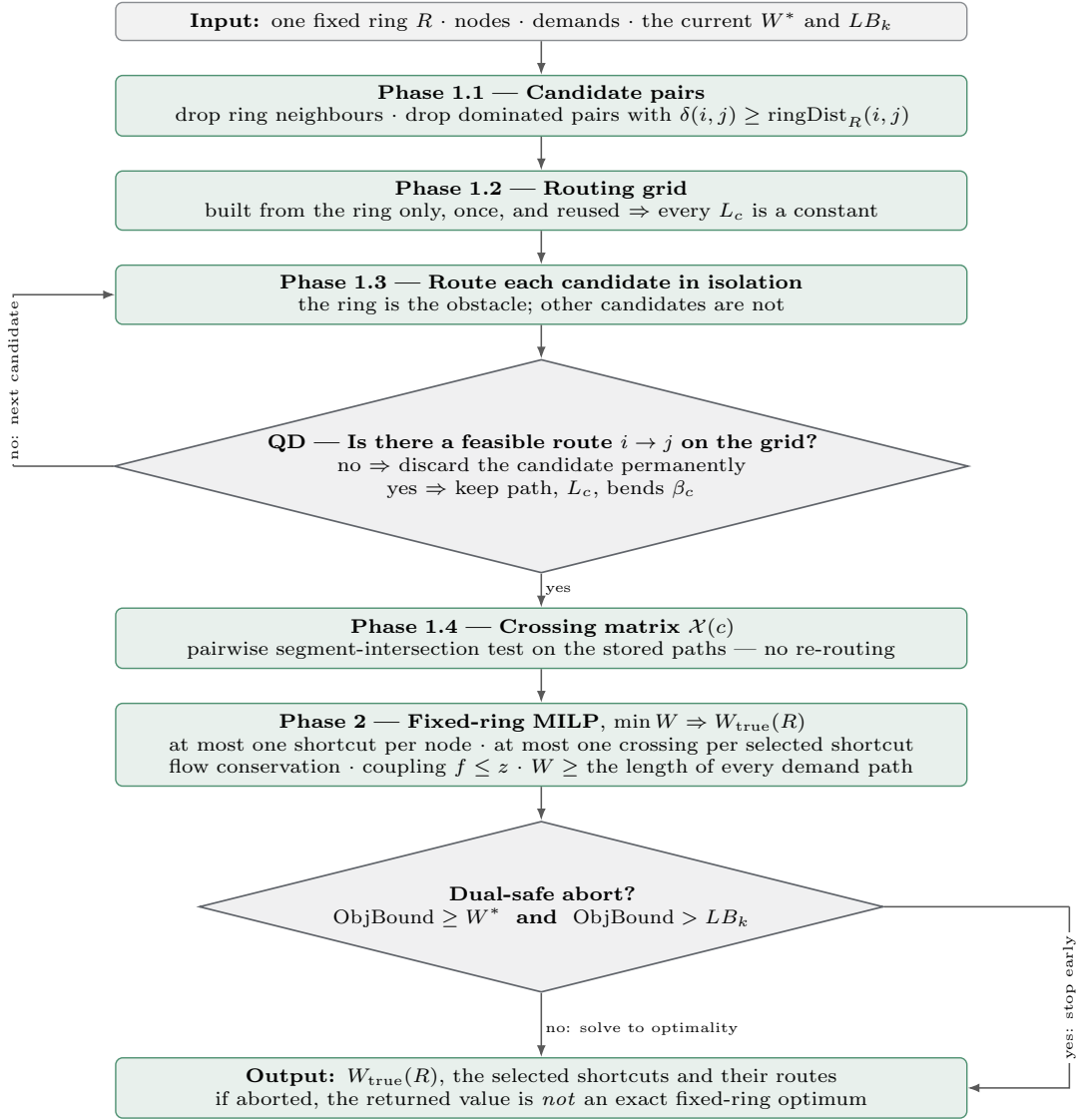
How to read this. The master is a *proxy*: a shortcut here is only a pair of endpoints charged its straight Manhattan distance $\delta(c)$, not a routed path. Always-crossing pairs may be selected together — each selected shortcut simply may have at most one such partner, the same physical rule the Shortcut Solver enforces on real paths. The dashed red box lists what is deliberately left out; that omission is precisely why LB_k is a lower bound and why every proposed ring still has to be verified.

4 Diagram 3 — Inside the Quick Estimator



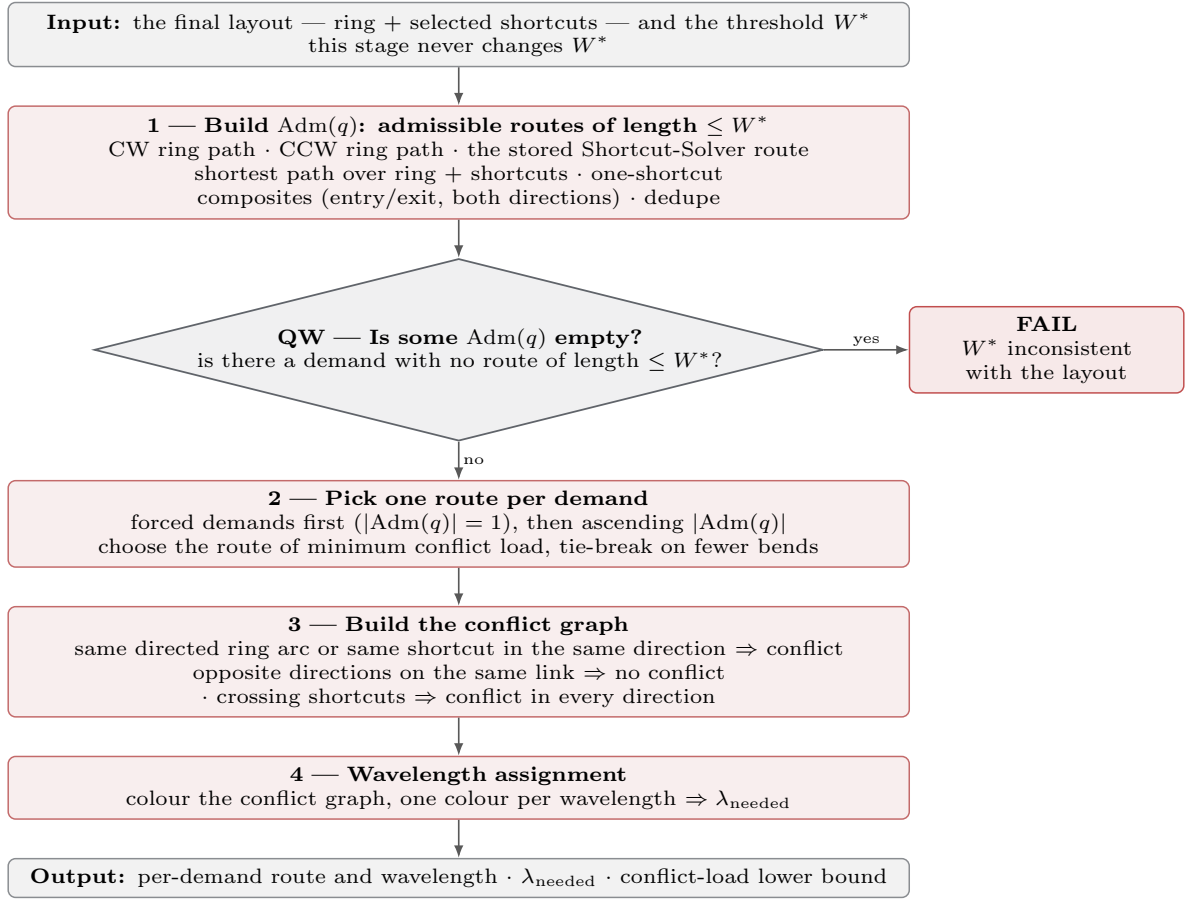
How to read this. This is the cheap middle step between a ring proposal and an expensive verification. Nothing is routed in the plane: the estimator only asks, per selected pair, whether a ring segment cuts straight across the rectangle between its endpoints. If not, the pair keeps $\delta(c)$; if so, a detour is unavoidable and a penalty derived from the overhang of the blocking segment is added. If every $\widehat{W}(R)$ were replaced by a random number the algorithm would still terminate with the same proven answer — only after more Shortcut Solver calls. That is why $\widehat{W}(R)$ enters no bound, no gap and no certificate.

5 Diagram 4 — Inside the Shortcut Solver



How to read this. The decisive structural fact sits in Phase 1.2: the grid depends on the ring alone, never on which shortcuts are chosen. Every candidate's length is therefore a constant computed once, and Phase 2 reduces to a purely combinatorial selection over pre-measured edges — other shortcuts affect only whether a combination is admissible, never how long a shortcut is. The abort is dual-safe because both inequalities are required: if the solver's own lower bound can no longer beat W^* and has already risen above LB_k , this ring can neither improve the incumbent nor produce a direct match, so stopping costs nothing. A run that was aborted must not be read as an exact fixed-ring optimum.

6 Diagram 5 — Wavelength postprocessing



How to read this. $\text{Adm}(q)$ is the menu of length-feasible routes for demand q ; QW is a genuine validation gate, not a formality, because an empty menu means the threshold and the layout contradict each other. The conflict rules are deliberately not uniform: for reuse of the same physical link — one ring arc, or one shortcut — opposite directions do not conflict, because ring and shortcut each consist of two independent directional waveguides. For two different shortcuts that cross, the conflict holds in all direction combinations, because the coupling happens at the crossing point regardless of which way light travels. Step 4 colours that graph; each colour is one wavelength.

7 Which quantity may do what

Quantity	Type	Updates W^* ?	Enters gap?	Can certify?
LB_k, LB_{k+1}	lower bound	no	yes	yes, together with W^*
$\widehat{W}(R)$	ranking heuristic	no	no	no
$W_{\text{true}}(R)$	feasible value	yes	via W^*	yes, direct match
W^*	upper bound	is the incumbent	yes	yes, bound catch-up
λ_{needed}	postprocess artefact	no	no	no

Removing the Quick Estimator would leave the algorithm correct and only make it slower, because verification effort would be spent in a worse order. Removing either bound would remove the ability to prove anything at all.